

Math Made Clear

Optional Subtitle

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About the Author

Short author bio here. (e.g., background and inspiration; include website/contact if desired.) Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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감사의 말 간단한 감사 인사나 크레딧

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저작권 모든 권리 보유. 이 책의 어떤 부분도 사전 허가 없이 복제하거나 사용할 수 없습니다.

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Preface

Mathematics is a universal language, but learning it in a second language can be both a challenge and an opportunity. This book was created especially for Korean students who are studying math in English. Our purpose is to bridge the gap between two worlds—helping you master essential math concepts while also building your confidence in English.

수학은 보편적인 언어이지만, 제2언어로 수학을 배우는 것은 도전이면서 동시에 기회가 될 수 있습니다. 이 책은 영어로 수학을 공부하는 한국 학생들을 위해 특별히 만들어졌습니다. 우리의 목적은 두 세계 사이의 간극을 메우는 것입니다—필수적인 수학 개념을 마스터하면서 동시에 영어에 대한 자신감을 키우는 것을 돕는 것입니다.

The motivation for this book comes from the experiences of many students who have struggled to understand math vocabulary, instructions, and problem-solving strategies in English. We know that with the right support, every student can succeed. By providing clear explanations, bilingual vocabulary, and step-by-step examples, we hope to make math accessible and enjoyable for you.

이 책의 동기는 영어로 된 수학 용어, 지시사항, 문제 해결 전략을 이해하는 데 어려움을 겪은 많은 학생들의 경험에서 비롯됩니다. 우리는 적절한 지원이 있으면 모든 학생이 성공할 수 있다는 것을 알고 있습니다. 명확한 설명, 이중 언어 어휘, 단계별 예시를 제공함으로써, 우리는 수학을 접근하기 쉽고 즐겁게 만들어 드리기를 희망합니다.

Whether you are preparing for international exams, planning to study abroad, or simply want to strengthen your math and language skills, this book is here to support your journey. Remember, every great mathematician started with curiosity and perseverance. We hope this book inspires you to explore, ask questions, and discover the beauty of mathematics in both Korean and English.

국제 시험을 준비하거나, 해외 유학을 계획하거나, 단순히 수학과 언어 능력을 강화하고 싶은 간에, 이 책은 여러분의 여정을 지원하기 위해 여기에 있습니다. 기억하세요, 모든 위대한 수학자는 호기심과 끈기로 시작했습니다. 우리는 이 책이 여러분이 탐구하고, 질문하고, 한국어와 영어로 된 수학의 아름다움을 발견하도록 영감을 주기를 희망합니다.

Introduction

Welcome to this math book designed especially for Korean students learning English! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

영어를 배우는 한국 학생들을 위해 특별히 설계된 이 수학책에 오신 것을 환영합니다! 이 책은 여러분이 영어 실력을 향상시키면서 동시에 수학에 대한 강한 기초를 쌓을 수 있도록 구성되어 있습니다. 각 섹션은 핵심 주제를 다루며, 명확한 설명으로 시작하여 예시, 연습 문제, 유용한 팁으로 이어집니다.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Geometry, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

이 책은 수와 연산, 식과 방정식, 기하학, 데이터와 통계 등의 주요 섹션으로 나뉩니다. 각 섹션 내에서는 단계별 가이드, 어휘 목록, 실제 응용 사례를 찾을 수 있습니다. 특별한 기능으로는 이중 언어 용어집, 피해야 할 일반적인 실수, 이해도를 확인할 수 있는 미니 퀴즈가 포함되어 있습니다.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with yourself and celebrate your progress!

이 책을 최대한 활용하려면, 각 섹션을 주의 깊게 읽고, 예시를 직접 시도해보며, 어휘 목록을 사용하여 수학 영어를 강화하는 것을 권장합니다. 복습이 필요하면 이전 섹션을 다시 방문하는 것을 주저하지 마세요. 기억하세요, 새로운 언어로 수학을 배우는 것은 시간과 연습이 필요합니다—자신에게 인내심을 가지고 진전을 축하하세요!

We hope this book becomes a valuable companion on your journey to mastering math and English together.

우리는 이 책이 여러분이 수학과 영어를 함께 마스터하는 여정에

서 소중한 동반자가 되기를 희망합니다.

Chapter 1

Numbers & Operations

Welcome! In this book, you'll explore the world of numbers and how we use them every day. You'll learn about place value, decimals, fractions, and more. Each Section includes explanations, examples, and practice problems to help you master these important math skills.

1.1 Place Value

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together.

We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

“In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number.”

1.1.1 What is Place Value?

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

To truly grasp the concept of place value, for example, let's take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit's value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let's see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In

another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

“Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.”

1.2 Decimals and Fractions

Sometimes, a whole number is too much. We want to describe something in parts: half a sandwich, a third of a mile, seventy-five hundredths of a dollar. That's when we use fractions and decimals. There are two ways of saying the same thing: less than one. Fractions use numerators and denominators. Decimals use place value. But both express parts of a whole.

Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$. The fraction $\frac{3}{10}$ becomes 0.3. The decimal 0.6 becomes $\frac{6}{10}$, which is $\frac{3}{5}$. Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.

“Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$.”

1.2.1 What are Decimals and Fractions?

As we move beyond whole numbers, we encounter quantities that are less than one. These are often represented as fractions or decimals. A **fraction** shows a part of a whole, such as $\frac{1}{2}$ or $\frac{3}{4}$. On the other hand, a **decimal** expresses the same idea using a decimal point, like 0.5 or 0.75. Both forms are just different ways to describe the same concept.

For example, consider the decimal 0.75. This can be written as the fraction $\frac{75}{100}$, which simplifies to $\frac{3}{4}$:

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Let's try converting between these forms. If you have the fraction $\frac{3}{10}$, you can write it as the decimal 0.3. Conversely, if you start with the decimal 0.6, you can express it as a fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$ in simplest form. Being able to move between decimals and fractions is a key skill that will help you in many areas of math.

“Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.”

1.3 Comparing and Ordering Numbers

Math is not just about knowing what numbers are. It is about knowing how they relate. We compare. We rank. We ask, which is bigger? Which is smaller? Which is equal? Symbols like $>$, $<$, and $=$ help us answer. But first, we have to understand the values.

Take 5.03 and 5.3. They may look close. But 5.3 is larger. Now try comparing fractions: $\frac{2}{5}$ and $\frac{3}{7}$. Convert them to decimals. $\frac{2}{5}$ is 0.4. $\frac{3}{7}$ is about 0.428. So $\frac{3}{7}$ is greater. This kind of comparison matters in real life—from prices to measurements to data.

Ordering numbers builds the same skill. Imagine you are given 0.2, $\frac{1}{3}$, 0.25, and 0.199. Which is the greatest? Which is least? Convert, compare, and you find the order. This process teaches more than math. It teaches precision, logic, and clarity.

“Math is not just about knowing what numbers are. It is about knowing how they relate.”

1.3.1 How do we compare and order numbers?

In mathematics, it's important not only to understand numbers, but also to compare them and put them in order. We use symbols like $>$ (greater than), $<$ (less than), and $=$ (equal to) to show how numbers relate to each other. For example, if you are given two numbers, you can decide which is larger or smaller, or if they are the same.

Let's look at a few examples. Suppose you want to compare 5.03 and 5.3. Since $5.03 < 5.3$, we use the less than symbol. Now, what if you have two fractions, $\frac{2}{5}$ and $\frac{3}{7}$? By converting them to decimals, you find that $\frac{2}{5} = 0.4$ and $\frac{3}{7} \approx 0.428$, so $0.4 < 0.428$, which means $\frac{2}{5} < \frac{3}{7}$.

Ordering numbers is also a useful skill. Imagine you are asked to arrange the numbers 0.2, $\frac{1}{3}$, 0.25, 0.199 from greatest to least. By converting them all to decimals, you see that $\frac{1}{3}$ (approximately 0.333) is the largest, followed by 0.25, then 0.2, and finally 0.199. Sometimes, you may need to compare a fraction and a decimal, such as $\frac{4}{9}$ and 0.44. Since $\frac{4}{9} \approx 0.444$, it is slightly greater than

0.44. These skills are essential for working with numbers in real-life situations.

“Take 5.03 and 5.3. They may look close. But 5.3 is larger.”

1.4 Factors and Multiples

Some numbers fit together cleanly. Others do not. Factors and multiples explain why. A factor is a number that divides evenly. A multiple is what you get when you multiply. These two ideas help us break numbers apart and build them back up.

If you list the factors of 12, you get 1, 2, 3, 4, 6, and 12. Each divides 12 without a remainder. The multiples of 5 are 5, 10, 15, 20, and so on. These ideas may seem mechanical, but they are powerful. They lay the groundwork for division, fractions, and algebra.

“A factor is a number that divides evenly. A multiple is what you get when you multiply.”

1.4.1 What are Factors and Multiples?

As you continue your journey in mathematics, you'll often need to break numbers down or build them up. This is where factors and multiples come in. A **factor** is a number that divides another number exactly, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12, because each of these divides 12 evenly. On the other hand, a **multiple** is what you get when you multiply a number by an integer. For instance, the multiples of 5 are 5, 10, 15, 20, and so on.

Let's see how this works with a few examples. If you want to find all the factors of 24, you list out 1, 2, 3, 4, 6, 8, 12, and 24. If you're asked for the first 6 multiples of 8, you simply multiply: 8, 16, 24, 32, 40, and 48. Sometimes, you might wonder if one number is a factor of another. For example, is 9 a factor of 81? Since $81 \div 9 = 9$ with no remainder, the answer is yes. Understanding factors and multiples is crucial for topics like division, fractions, and finding patterns in numbers.

1.5 Prime and Composite Numbers

Some numbers stand alone. Others are made of parts. Prime numbers have only two factors: 1 and themselves. Composite numbers have more. This is not just a label—it is a clue to how numbers work.

2, 3, 5, 7, 11, 13, and 17 are prime. They cannot be broken down. But 15 is composite: it splits into 3 and 5. Learning to recognize prime and composite numbers helps us factor numbers, simplify fractions, and understand the building blocks of math.

“Prime numbers have only two factors: 1 and themselves.
Composite numbers have more.”

1.5.1 What are Prime and Composite Numbers?

As you explore numbers further, you’ll discover that some numbers have only two factors, while others have many. A **prime number** is a number greater than 1 that has exactly two distinct factors: 1 and itself. For example, 2, 3, 5, 7, 11, 13, 17, and 19 are all prime numbers under 20. In contrast, a **composite number** has more than two factors. For instance, 15 is composite because it can be divided by 1, 3, 5, and 15.

Let’s practice identifying these. Consider the numbers 14, 17, 21, 23, and 25. Among these, 17 and 23 are prime because they have only two factors. The others—14, 21, and 25—are composite. If you look at all the composite numbers between 10 and 20, you’ll find 10, 12, 14, 15, 16, 18, and 20. It’s also important to note that the number 1 is neither prime nor composite, since it has only one factor: itself. Recognizing prime and composite numbers is a foundational skill for understanding more advanced math concepts like prime factorization and divisibility.

1.6 GCM & LCM

Sometimes two numbers meet in the middle. The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple. These ideas sound opposite, but they both help us solve problems with shared values.

Take 12 and 18. Their GCF is 6 since it is the biggest number that divides them both. Their LCM is 36 which is the smallest number they both fit into. We find these using prime factorization. Once you learn how, you see how numbers connect. And once you see the connection, math becomes clearer.

“The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple.”

1.6.1 What are GCF and LCM?

When working with two or more numbers, it's often useful to find numbers they have in common. The **Greatest Common Factor (GCF)** is the largest number that divides two numbers exactly, while the **Least Common Multiple (LCM)** is the smallest number that both numbers divide into evenly.

To find the GCF and LCM, you can use prime factorization. For example, let's find the GCF and LCM of 12 and 18. The prime factorization of 12 is $2^2 \times 3$, and for 18 it's 2×3^2 . The GCF is the product of the lowest powers of common primes: $2^1 \times 3^1 = 6$. The LCM is the product of the highest powers: $2^2 \times 3^2 = 36$.

Let's apply this to a few more examples. The GCF of 24 and 36 is 12, since 12 is the largest number that divides both. The LCM of 8 and 12 is 24, which is the smallest number both can divide into. For 14 and 28, we see that $14 = 2 \times 7$ and $28 = 2^2 \times 7$, so the GCF is 14 and the LCM is 28. Mastering GCF and LCM will help you with fractions, ratios, and solving many real-world problems.

1.7 Tip for Students

Math is not just about solving problems. It is about seeing how ideas fit together. Learn the basics—place value, fractions, comparisons, factors, primes, and multiples. They are not just chapters. They are tools. And the better you understand them, the more confidently you can use them anywhere.

Interesting Facts

- The word “digit” comes from the Latin word for finger, because people used their fingers to count.
- The decimal system is called “base-10” because each place is ten times the value of the place to its right.
- The largest single-digit number in the decimal system is 9.
- The introduction of zero by Indian mathematicians made calculations much clearer and easier.

Example sentence is here. (Kim 2003, p. 359).

“Example sentence is here.” The author said this (*Help your kids with math* 2010, p. 200).

1.8 Additional Concepts

This section covers essential vocabulary and techniques that will help you master the fundamental concepts of numbers and their relationships. These additional tools complement the main concepts and provide practical methods for solving problems.

1.8.1 Place Value Vocabulary

Understanding place value requires familiarity with specific terminology. The essential place values you need to learn include billions, hundred millions, ten millions, millions, hundred thousands, ten thousands, thousands, hundreds, tens, and ones.

“한국어와 비슷하게 앞자리에 10배만 붙이면 됨: 일 십 백 천 만 십만 백만 천만 억 십억 백억”

“ones(일), tens(십), hundreds(백), thousands(천), ten thousands(만) 영어는 천부터 $\times 10$ ”

For decimal place values, the key terms are tenths, hundredths, thousandths, ten-thousandths, and millionths.

“똑같은 건데 뒤에 th만 추가”

“반드시 알아야하는 건 tenths(할), hundredths(푼), thousandths(리)”

“이외의 것은 굳이”

1.8.2 Fraction Fundamentals

Fractions represent parts of a whole, and understanding their components is crucial. Every fraction has two main parts: the numerator and the denominator.

“denominator(분모), numerator(분자)”

“외우기 힘들면 그냥 AND로 외우면 됨 and 순서대로 n/d ”

There are different types of fractions that serve different purposes. An improper fraction has a numerator larger than its denominator, while a mixed number combines a whole number with a proper fraction.

“improper(가분수), mixed(대분수)”

“이건 솔직히 그냥 성질만 알면 바로 외워짐; 가분수는 대가리가 크니까 당연히 비율적으로 부적절(improper)함; 대분수는 자연수와 분수의 혼합(mix)”

1.8.3 Decimal Types

Decimals come in different forms, each with unique characteristics. A repeating or recurring decimal has a pattern that continues infinitely, while a finite decimal has a limited number of digits after the decimal point. An infinite decimal continues without end.

“repeating/recurring(순환소수), finite(유한), infinite(무한)”

“repeating/recurring(반복), in-(없다) + finite(제한이 있는) = infinite(무제한)”

“순환소수의 구별방법은 소수점위에 점의 유무 (ex1: 1.3-위에 점 찍혔다 치면 1.333333333; ex2: 1.3475 3과 5위에 점이 찍혔을 경우 그 구간이 순환 1.347534753475)”

“무한소수는 분수로의 변환이 불가하다”

“순환소수의 분수 변환은 변환구간을 9로 나눈다 (ex: 1.3984에서 3984가 순환할 경우 $3984/9999 = 0.398439843984 \dots$)”

1.8.4 Comparison Vocabulary

Mathematics uses specific language to describe relationships between numbers. Understanding these terms helps you interpret problems correctly and communicate mathematical ideas clearly.

For greater than relationships, we use terms like greater than, more than, larger than, longer than, higher than, above, exceeds, and over. For greater than or equal to, we say greater than or equal to, at least, more than or equal to, minimum, no less than, not under/below, or not smaller than.

For less than relationships, we use less than, fewer than, smaller than, shorter than, lower than, under, below, or beneath. For less than or equal to, we say at most, does not exceed, maximum, no more than, up to, or not above/greater than.

“좀 많긴한데 at least/minimum 이랑 at most/maximum 만
외우면 sat 수학은 해결”

“무작정 외우지 말고 문제 읽을때 한번 뜻을 생각해 보
면 이해가 됨”

1.8.5 Prime Factorization

Prime factorization is the process of breaking down a number into its smallest building blocks—prime factors. This technique is fundamental for understanding number structure and solving problems involving factors and multiples.

“prime factorization(소인수분해) 말그대로 prime(소
수)까지 분해한다는 거임”

“제일 기본적인 순서는 소인수분해할 숫자를 보는거
임”

“그 숫자가 몇의 배수인지 봐야하는데 이걸”

To determine if a number is divisible by certain factors, we can use these rules:

- 2의 배수: 짝수/일의 자리 숫자가 2, 4, 6, 8, 0인 수
- 3의 배수: 각 자리의 수의 합이 3의 배수
- 끝의 두 자리 수가 00이거나 4의 배수
- 5의 배수: 일의 자리 숫자가 0또는 5
- 7의 배수: 세자리수 abc 일 때 $ab-c*2$ 가 7의 배수인수 (이건 솔직히 몰라도 됨)

“일단 대충 저걸로 판단후 최소 값부터 계속 소인수분해 (ex: 60을 소인수 분해 하면 일단 2로 나눈다 왜냐 하면 짝수이기 때문; $2*30$; 30도 짝수이기 때문에 2로 다시 나눈다; $2*2*15$; 15는 일의 자리가 5이기 때문에 5로 나눈다; $2*2*5*3$; 와 다 소수네; 소인수분해 완료)”

1.8.6 Finding GCF and LCM

The greatest common factor (GCF) and least common multiple (LCM) are essential concepts for working with multiple numbers. While these ideas may seem complex, there's a simple method using the Korean "ㄴ" technique that makes finding them straightforward.

“한국 학교에서 당연히 배웠겠지만 'ㄴ'자로 하면됨”

“'ㄴ'자 방법이란 소인수분해랑 똑같은 건데 두수 혹은 그 이상의 숫자들이 공유하는 약수로 다 나눈 건데 이거는 꼭 소수로 안해도 됨”

Let's find the GCF of 12 and 18 using this method:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ \hline & 2 & 3 \end{array}$$

“위에처럼 마지막 숫자가 소수가 되면 끝”

“최대공약수의 경우 'ㄴ'자 옆에 있는 숫자들의 곱; 지금 같은 경우는 6”

Now let's find the LCM of 12 and 18:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ \hline & 2 & 3 \end{array}$$

“최소공배수의 경우 'ㄴ'자 옆 밑에 있는 모든 숫자들의 곱; 지금 같은 경우는 $6 \cdot 2 \cdot 3 = 36$ ”

“와 쉽죠”

These methods provide efficient ways to work with factors and

multiples, making complex problems more manageable and helping you develop a deeper understanding of number relationships.

Appendix

Important Formulas and Definitions

- **Place Value:** Value of a digit = digit $\times$ place value.
- **Fraction to Decimal:** $\frac{a}{b} = a \div b$.
- **Prime Number:** A number greater than 1 with exactly two factors: 1 and itself.
- **Composite Number:** A number with more than two factors.
- **GCF:** Product of lowest powers of common primes.
- **LCM:** Product of highest powers of all primes present.

Further Reading and Resources

- "Elementary Mathematics" by Liping Ma
- "How to Be a Math Genius" by Mike Goldsmith
- Khan Academy: <https://www.khanacademy.org/>
- Purplemath: <https://www.purplemath.com/>

Summary of Key Concepts

Topic	Key Points
Place Value	Each digit's value depends on its position. Example: $4275 = 4000 + 200 + 70 + 5$.
Decimals and Fractions	Decimals and fractions both represent parts of a whole. Example: $0.75 = \frac{3}{4}$.
Comparing and Ordering Numbers	Use $>$, $<$, $=$ to compare. Convert to decimals for easy comparison. Example: $\frac{2}{5} < \frac{3}{7}$.
Factors and Multiples	Factors divide a number evenly; multiples are products of a number and an integer. Example: Factors of 12 are 1, 2, 3, 4, 6, 12.
Prime and Composite Numbers	Primes have exactly two factors (1 and itself); composites have more. Example: 2, 3, 5 are prime; 4, 6, 8 are composite.
GCF and LCM	GCF: largest common factor; LCM: smallest common multiple. Use prime factorization. Example: GCF of 12 and 18 is 6; LCM is 36.

Table 1.1: Summary of Key Concepts

References

- [1] Sŏn-ju Kim. *SSAT-SAT suhak yong'ŏ sajŏn: yŏng'ŏ ro paeu nŭn suhak*. kor. Ch'o-pan. OCLC: 52616988. Sŏul-si: Ez-book, 2003. ISBN: 9788956240510.
- [2] *Help your kids with math: a unique step-by-step visual guide*. First American edition. New York, New York: DK Publishing, 2010. ISBN: 9780756649791.